

Exercise Set 3

Quantitative Trading and Price Impact
MSc in Mathematics and Finance

2025-2026

- (a) **Fitting price impact models.** Fix a half-life $H > 0$, a model type (linear or square-root), and an explanation horizon $\Delta \in \{1, 5, 10, 30\}$ minutes. Work on the 10-second grid, and let $\tau := \Delta/(10\text{s})$ be the corresponding number of time bins.

We model the relative price changes as

$$\frac{P_t - P_{t-\tau}}{P_{t-\tau}} = \lambda (\bar{I}_t - \bar{I}_{t-\tau}) + \text{noise},$$

where I_t is the impact state, $\bar{I}_t := I_t/\lambda$ is the normalized impact state, and $\lambda > 0$ is the impact coefficient. This motivates a linear regression of returns on changes in the normalized impact state, with the regression slope serving as an estimate of λ .

Let P_t denote the midprice and let I_t be the intraday impact state computed as in Exercise Set 2(b). For all bins $t \geq \tau$, define

$$x_t := \bar{I}_t - \bar{I}_{t-\tau} \quad \text{and} \quad y_t := \frac{P_t - P_{t-\tau}}{P_{t-\tau}}.$$

That is, x_t is the change in normalized impact state over the horizon Δ , and y_t is the corresponding return over the same horizon.

For each calendar month and each symbol, fit the pooled linear regression

$$y_t = \alpha + \beta x_t + \varepsilon_t,$$

using all trading days in that month and all intraday bins (optionally restricting to a time window such as $t \geq 10:00$ to avoid start-of-the-day effects).

Interpret the fitted slope β as an estimate of the impact coefficient λ , and report (α, β) per symbol and month. Finally, evaluate the in-sample R^2 on that month as well as the out-of-sample R^2 when applying the fitted coefficients to the subsequent month.

- (b) **Computing markouts.** Unzip and load the fill sample data.
- (i) For a single stock on a single trading day, let $(v_i, p_i, t_i)_{i=1}^N$ denote the signed fill volume, fill price, and fill time of each fill, and let P_t denote the midprice process at fill timestamps. Fix a horizon $h > 0$. The markout of fill i is defined by

$$\left(\frac{P_{t_i+h} - P_{t_i}}{P_{t_i}} \right) \text{sgn}(v_i).$$

Implement a function that computes one markout per fill.

Hint: To obtain P_{t_i+h} for each fill time t_i , create a shifted copy of the fill price series indexed by $t_i - h$ and use `merge_asof` to align each fill at time t_i with the most recent observation of the shifted series.

- (ii) For AAPL stock and each trading day, compute per-fill markouts and then compute the daily average markout (mean over fills) for the horizons

$$h \in \{10s, 1min, 5min, 30min, 1h, 2h\}.$$

Create a table with one row per date and one column per horizon. Then merge into this table two additional daily covariates from Exercise Set 2(a): the daily (10-second) price volatility and the daily traded volume.

- (iii) For AAPL stock and each horizon h , fit the linear regression

$$\text{markout}_h = c_h + a_h \text{volatility} + b_h \text{volume} + \varepsilon,$$

and report the estimated coefficients and R^2 .